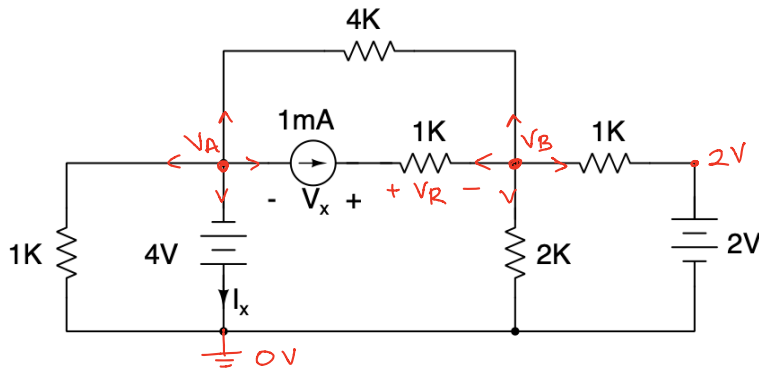


Duration: 20 minutes

For the circuit shown below:

- (10 pts) Select a common node and mark it as the ground node. Label the other node voltages in the circuit as $V_A, V_B, \dots$
- (50 pts) Find the node voltages using node analysis.
- (20 pts) Find the current I_x as marked in the circuit.
- (20 pts) Find the voltage V_x as marked in the circuit.



a) V_A and V_B are marked on the circuit.

b) $V_A = -4V$

KCL at B:
$$\frac{V_B - (-4)}{4K} - 1mA + \frac{V_B}{2K} + \frac{V_B - 2}{1K} = 0$$

$$(4K) \quad (2) \quad (4)$$

$$7V_B = 8 \Rightarrow V_B = \frac{8}{7} V$$

c) To find I_x , write KCL at node A:

$$I_x + \frac{V_A}{1K} + 1mA + \frac{V_A - V_B}{4K} = 0$$

$$I_x - \frac{4}{1K} + 1mA + \frac{-4 - \frac{8}{7}}{4K} = 0 \Rightarrow I_x = \frac{30}{7} mA$$

d) From KVL:

$$V_B - V_A = -V_R + V_x$$

$$V_x = V_B - V_A + V_R = \frac{8}{7} + 4 + (1mA \times 1K) = \frac{43}{7} V$$